

The Learning Loop - GNN for Klebsiella Drug Discovery

Project Overview

This project uses a Graph Neural Network (GNN) model to predict whether a chemical compound may act as an inhibitor of the Klebsiella SSB-PriA protein target.

Features include:

- Single compound prediction using SMILES strings
 - Bulk prediction via CSV upload
 - Molecule visualization
 - Prediction history
 - Analytics dashboard
 - CSV export of prediction results
-

Prerequisites

Before running the project, install:

- Python 3.10 or newer
 - Git
 - VS Code (recommended)
-

Clone Repository

```
git clone <repository-url>
cd TheLearningLoop
```

Create Virtual Environment

Windows:

```
python -m venv venv
```

Activate virtual environment:

```
venv\Scripts\activate
```

Mac/Linux:

```
python3 -m venv venv  
source venv/bin/activate
```

Install Dependencies

Install all required packages:

```
pip install -r requirements.txt
```

Database Setup

The project automatically creates and uses a SQLite database.

Verify the database folder exists:

```
database/
```

The application will create the required tables if they do not already exist.

Required Folders

Ensure the following folders exist:

```
uploads/  
molecule_images/  
database/
```

If missing, create them manually.

Start Backend API

From the project root:

```
uvicorn backend.app:app --reload
```

Backend API runs at:

```
http://127.0.0.1:8000
```

Verify the API is running:

```
http://127.0.0.1:8000/docs
```

Swagger documentation should appear.

Start Frontend

Open:

```
frontend/index.html
```

Recommended:

Use VS Code Live Server extension.

OR

Serve using Python:

```
cd frontend  
python -m http.server 8080
```

Open:

```
http://localhost:8080
```

Features Available

Analyze

- Enter a SMILES string
- Click Analyze Compound
- View prediction result
- View generated molecule image

Bulk Prediction

- Upload a CSV file containing SMILES strings
- Run batch prediction
- View results table
- Download prediction results as CSV

CSV format:

```
smiles  
CCO  
CCN  
CCC
```

History

- View previous predictions
- Paginated results
- Displays:
 - SMILES
 - Prediction
 - Confidence

- Timestamp
-

Analytics

Dashboard includes:

- Total predictions
 - Active predictions
 - Inactive predictions
 - Prediction distribution chart
 - Top searched compounds chart
-

Project Structure

TheLearningLoop/

```
|— backend/
|   |— app.py
|   |— database.py
|   |— predictor.py
|   └— tests/
```

```
|— frontend/
|   |— index.html
|   |— css/
|   |— js/
|   └— views/
```

```
|— database/
|— uploads/
|— molecule_images/
|— requirements.txt
└— README.md
```

Running the Application

1. Activate virtual environment
2. Install requirements
3. Start FastAPI backend

```
uvicorn backend.app:app --reload
```

1. Open frontend

```
frontend/index.html
```

or

```
http://localhost:8080
```

1. Use Analyze, History, and Analytics tabs

Troubleshooting

API Not Running

Verify:

```
uvicorn backend.app:app --reload
```

is running without errors.

Molecule Images Not Displaying

Verify:

```
molecule_images/
```

exists and is writable.

CSV Upload Fails

Verify:

```
uploads/
```

exists and contains write permissions.

Python Package Errors

Reinstall requirements:

```
pip install -r requirements.txt
```

Notes

- Frontend and backend run separately.
- Backend must be running before using the frontend.
- Prediction history is stored in the local SQLite database.
- Analytics data is generated from stored prediction history.